

Soham Belurgikar

Champaign, IL | +1 (217) 402-1504 | sohamsb2@illinois.edu | linkedin.com/in/sohamb17 | github.com/sohamb17 | sohamb17.github.io

EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science (MCS)

Sardar Patel Institute of Technology - University of Mumbai

B.Tech in Computer Engineering; CGPA: 9.21/10.0

Urbana-Champaign, IL

Aug. 2026 – May 2028 (Expected)

Mumbai, India

Aug. 2019 – May 2023

EXPERIENCE

Barclays

Software Engineer (BA4)

Pune, India

Apr. 2025 – Jul. 2026

- Built a distributed comparison framework for 10+ node Spark clusters, isolating race conditions and cutting regression verification time across runtime versions by 50%.
- Architected an on-demand synthetic data platform using React, Spring Boot, and Python, generating 1M+ records to enable load testing without production PII.
- Automated data reconciliation and validation pipelines across staging and production, reducing manual verification effort by approximately 40%.

Software Engineer (BA3)

Jul. 2023 – Mar. 2025

- Modernized core distributed data pipelines by migrating batch workloads to Spark 3.x on open-source Hadoop (HDFS), eliminating runtime bottlenecks with zero production downtime.
- Engineered automated CI/CD pipelines via Jenkins and Groovy, cutting deployment cycle times by 35% across 15+ distributed artifacts and staging environments.
- Benchmarked and tuned Spark SQL query plans across 10+ TB financial datasets, resolving execution bottlenecks while preserving 100% data integrity.

Summer Intern (BA3)

Jun. 2022 – Jul. 2022

- Developed a Python tool that automated cross-system dataset reconciliation and generated structured discrepancy reports for millions of records, packaged in Docker and integrated into a Jenkins pipeline, enabling the team to spot data mismatches quickly and cut manual review effort.

PROJECTS

FrameSeek – Multimodal Video Search Engine

[GitHub](#) | [Demo](#)

Python, Go, FastAPI, React, TypeScript, PostgreSQL, pgvector, LangChain, LangGraph, Docker

- Built a multimodal video search engine that indexes speech (Whisper), on screen text (Tesseract OCR) and frames (OpenCLIP) into timestamped clips across five retrieval channels, with LangGraph conversational search over a custom LangChain retriever.
- Engineered resumable ingestion on PostgreSQL with job leases, attempt generations, checksummed stage manifests and atomically published index versions, verified by worker crash, duplicate submission and failed reindex tests.
- Developed a Go API with streaming uploads, SSRF safe URL import, idempotent jobs and signed media URLs in front of a Python FastAPI retrieval service on pgvector and PostgreSQL full text search.
- Trained a scikit-learn relevance scorer on 25 cross modal features that raised top 1 success on held out videos from 26% to 48% over fixed rank fusion, measured on 79 AI written labels with a random author audit.

StreamForge – Distributed Real-Time Feature Store & Pipeline

[GitHub](#) | [Live Demo](#)

Go, Apache Kafka, gRPC, Redis, PostgreSQL, Python, Docker, Prometheus, React

- Built a real time feature store in Go and Kafka computing per zone ML features from NYC taxi trips; sustained 250,000 synthetic events/sec for 10 minutes with flat consumer lag and 63 ms p99 freshness on a 2 vCPU host.
- Implemented Raft from scratch in Go for a 5 node online store with PreVote, ReadIndex, snapshots and an fsynced WAL; verified linearizability with Porcupine under leader kills, pauses and network partitions, with 381 ms median failover.
- Achieved effectively once processing by committing Kafka offsets and window state atomically with feature updates in each Raft entry, yielding 0 mismatches against a batch oracle across 1M trips under repeated worker crashes and duplicate redelivery.
- Served features over gRPC with deadlines at 13 ms p99 for linearizable reads under 1,000 req/s open loop load, and added request coalescing that cut Redis cached p99 at 2,000 req/s from 33 ms to 22 ms.

TECHNICAL SKILLS

Languages: Python, Go, Java, TypeScript, C++, SQL, C, Groovy, Bash

Distributed Systems & Data: Apache Spark, Apache Hive, Apache Kafka, Hadoop (HDFS), PostgreSQL, pgvector, Redis

ML & Retrieval: PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, pandas, NumPy

Cloud & DevOps: AWS (S3, Batch, EC2), Docker, Kubernetes, Jenkins, CI/CD, AI coding assistants (GitHub Copilot, Cursor, Claude Code, OpenAI Codex, Gemini), Linux, Git, GitLab

Frameworks & Protocols: Spring Boot, FastAPI, Django, React, Angular, Node.js, REST APIs, gRPC

PUBLICATIONS

[\[Google Scholar Profile\]](#)

- A Novel Approach to Fall Detection System using Computer Vision.** Co-author, ICACCS 2023.
- Street Smart: An Intelligent Traffic Management System.** Co-author, INDIACom 2023.